

Predictive Analytics on Large-Scale NYC Taxi Trip Data (January–June 2025)

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Abstract

This report details the implementation of an end-to-end Big Data analytics pipeline to predict trip outcomes for NYC Yellow Taxis using 24 million records from January–June 2025. Leveraging Apache Spark on Databricks, we addressed three predictive tasks: duration regression, fare regression, and binary trip-length classification. A comparative analysis of linear baselines and Random Forest ensembles was conducted. Results demonstrate that ensemble methods significantly outperform linear models in duration prediction ($R^2 = 0.81$ vs 0.69), while classification reached an ROC-AUC of 0.94 . The study highlights the efficacy of distributed processing and non-linear modelling in urban mobility contexts.

I. INTRODUCTION

Large-scale mobility datasets have become increasingly important for urban planning, transportation optimisation, and demand forecasting. Taxi trip records provide a rich source of information about travel behaviour, congestion patterns, and pricing dynamics in city environments. Predicting trip outcomes such as duration and fare is critical for fleet management and pricing optimization. The objective is to demonstrate a scalable pipeline that handles 24 million records, far exceeding traditional single-machine processing limits. This project investigates three primary hypotheses:

- **Hypothesis 1:** Trip duration can be accurately predicted using distance and temporal features.
- **Hypothesis 2:** Fare amounts follow a predictable linear relationship with distance and time.
- **Hypothesis 3:** Trips can be classified as short or long (≥ 15 mins) based on trip based features.

II. DATASET DESCRIPTION

The dataset consists of six months of yellow taxi records from January to June of 2025 sourced from the NYC Taxi & Limousine Commission

(TLC) website [1] as six parquet files (one for each month of data). **Data Volume:** The raw dataset initially contained 24,083,384 records, each containing 20 variables. **Variables:** Included fare, distances, timestamps, location IDs, tip and fee amounts, and payment types. **Data Quality:** Initial analysis via `df.describe()` and histograms identified significant noise, such as negative fare amounts and trips with zero distance as well as extreme outliers that required statistical pruning.

Feature	Description
trip_distance	Trip distance (miles)
passenger_count	Number of passengers
PULocationID	Pickup zone ID
DOLocationID	Drop off zone ID
fare_amount	Base fare charged
tpep_pickup_datetime	Pickup Timestamp
tpep_dropoff_datetime	Dropoff Timestamp

TABLE I
KEY FEATURES IN THE DATASET.

III. METHODOLOGY

The pipeline was implemented in Databricks using PySpark, ensuring distributed processing of the 24 million row dataset.

A. Data Ingestion & Cleaning

Data was ingested into the Databricks environment as Parquet files Spark's native Parquet reader

and merged into one Dataframe. For exploratory visualizations, a 1% sample was converted to a Pandas Dataframe to optimize memory usage while maintaining a representative distribution of trip distances and fares.

The cleaning process involved five distinct stages:

- 1) **Duplication Removal:** Use of `dropDuplicates()` to ensure unique trip records. No duplicates found.
- 2) **Validity Filtering:** Removal of records where `trip_distance`, `fare_amount`, or `passenger_count` were zero or negative.
- 3) **Temporal Pruning:** Trips with a duration ≤ 0.01 minutes were discarded as errors.
- 4) **Statistical Outlier Detection:** While a hard cut-off of \$400 was applied to fares, a **3-Sigma Rule** was implemented for distance and duration. Only records within three standard deviations of the mean were retained:

$$x < \mu + 3\sigma$$

This method effectively pruned extreme right-tail outliers (e.g., cross-state trips) that would otherwise skew the regression coefficients. Although taxi trip distributions are right-skewed, and the 3-Sigma rule assumes a normal distribution, the extremely large dataset size ensures that removing extreme outliers does not materially bias the underlying data distribution.

- 5) **Noise Reduction:** To reduce dimensionality and noise, 11 irrelevant columns (e.g., `RatecodeID`, `Airport_fee`) were dropped.

B. Feature Engineering

- **Trip Duration:** The `trip_duration` feature was derived using the difference between Unix timestamps of drop-off and pick-up times, converted to minutes. This is to allow models to capture daily demand cycles and congestion patterns that influence both trip duration and fare structure.
- **Timestamp granularity:** Using Spark functions timestamps were broken down into `pickup_hour`, `pickup_day` and `pickup_month` features. Post-extraction,

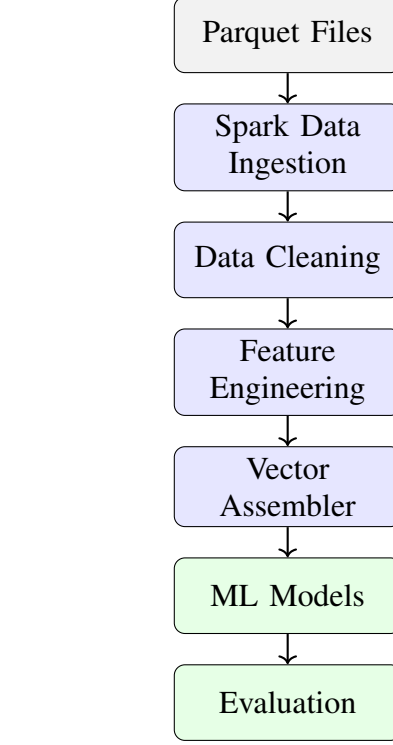


Fig. 1. Modern machine learning pipeline architecture implemented using Apache Spark in the Databricks environment. The pipeline performs ingestion, cleaning, feature engineering, and model training on large-scale taxi trip data.

the original `tpep_pickup_datetime` was dropped to reduce dimensionality.

- **Categorical Flags:** Two binary indicators were engineered to represent demand periods:
 - `is_weekend`: Flags trips occurring on Saturday (7) or Sunday (1).
 - `rush_hour`: Identified as 07:00–10:00 and 16:00–19:00, typically periods of increased morning and evening traffic.
- **Target Variable for Classification (`trip_class`):** To address **Hypothesis 3**, a binary label was generated using a threshold-based approach. Trips with a `trip_duration` of less than 15 minutes were classified as 'Short' (0), while trips over 15 minutes were 'Long' (1). The cleaned and feature enhanced dataset was persisted as a Delta Table for use in separate modelling notebooks, due to issues with databricks server-less caching overflow when attempting to run models in succession.

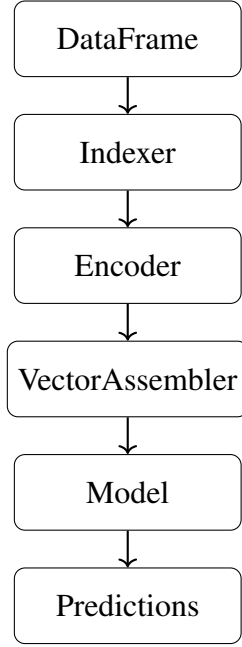


Fig. 2. ML pipeline architecture used for linear and logistic regression models.

C. Modelling Pipeline Architecture

Models were implemented using the `pyspark.ml.Pipeline` API. To ensure fairness in comparison, two distinct feature-processing paths were used:

- 1) **Linear Path:** Used `OneHotEncoder` and `StandardScaler` to handle categorical indices and ensure feature magnitudes did not skew the gradient descent.
- 2) **Forest Path:** Used `StringIndexer` and raw vectors, as tree-based models are invariant to feature scaling and can interpret categorical indices directly.

Due to memory constraints on the Databricks cluster, we utilized explicit variable deletion (`del`) and notebook modularization to prevent driver-node cache overflow.

D. Predictive Modeling Implementation

The modeling phase was structured as a series of supervised learning experiments designed to test the three primary hypotheses. To ensure scalability and reproducibility, all models were implemented using the `pyspark.ml.Pipeline` API.

1) *Regression for Duration and Fare (Hypothesis 1 & 2):* To address the predictability of trip duration and fare amounts, we compared a traditional linear approach against a modern ensemble method:

- **Linear Regression (LR):** This was selected as a baseline model for its high interpretability. LR assumes a linear, additive relationship between features like `trip_distance` and the target variables. It serves as a benchmark to measure the "linearity" of the NYC taxi ecosystem.
- **Random Forest Regressor (RFR):** Urban traffic patterns are inherently non-linear; the relationship between distance and time changes significantly during rushhour. RFR is an ensemble alternative capable of capturing these complex feature interactions. The model was configured with **50 trees** and a **maxDepth of 10** to provide robust generalization while managing computational costs on the distributed cluster.

2) *Binary Classification for Trip Length (Hypothesis 3):* To test if taxi trips can be categorized into "Short" vs. "Long" durations, we utilized the engineered `trip_class` label.

- **Logistic Regression:** This was chosen to evaluate the effectiveness of a linear decision boundary for binary classification, providing a probabilistic interpretation of class membership.
- **Random Forest Classifier (RFC):** RFC was employed to handle the high cardinality of the `PULocationID`. We utilized **300 maxBins** within the RFC, which is a critical technical choice. This setting allows the model to treat each unique taxi zone as a distinct split point, capturing location-specific traffic density that significantly impacts trip duration.

3) *Evaluation and Computational Management:* Each pipeline was evaluated on a held-out 20% test set using the `RegressionEvaluator` and `MulticlassClassificationEvaluator` classes. Due to the high volume of 24 million records, iterative model testing was conducted in modular notebooks. After each evaluation, explicit variable deletion (`del`) was performed to clear the driver node's cache, ensuring the stability of the Databricks environment and preventing memory overflow during the training of complex ensemble models.

IV. RESULTS AND ANALYSIS

A. Exploratory Data Analysis (EDA) Findings

Prior to modeling, a 1% sample of the 24-million-record dataset was analyzed. Key observations included:

- **Feature Correlation:** A strong linear correlation was observed between `trip_distance` and `fare_amount`. However, `trip_duration` exhibited a high degree of variance relative to distance, suggesting that external factors like traffic density play a critical role.
- **Data Quality:** The application of the 3-Sigma rule effectively removed non-physical outliers, resulting in a cleaner distribution for gradient-based learning.
- **Noise Reduction:** Though EDA's data cleaning, the number of rows was reduced from over 24 million to 17.69 million. Over a quarter of the data recorded was invalid, majority with some zero/negative values.

B. Duration Prediction (Hypothesis 1)

The comparison between Linear Regression and Random Forest (RF) for trip duration highlighted the limitations of linear assumptions in urban traffic modeling.

TABLE II
DURATION PREDICTION PERFORMANCE

Model	RMSE	MAE	R^2
Linear Regression	7.33	4.86	0.690
Random Forest	5.70	3.60	0.812

The RF model achieved an R^2 of **0.812**, a significant improvement over the 0.690 achieved by the baseline. Analysis of RF feature importance revealed that while `trip_distance` (0.678) is the primary driver, `PULocationID_index` (0.251) and `pickup_hour` (0.053) are vital for capturing non-linear congestion patterns.

C. Fare Prediction (Hypothesis 2)

Fare prediction yielded the highest accuracy across all regression tasks. This could be expected due to the regulated pricing structure of NYC taxi fares, where cost is primarily determined by distance and predefined tariff rules. [2]

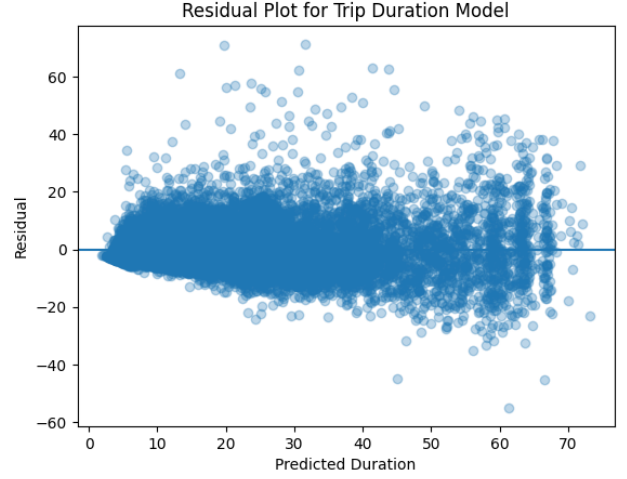


Fig. 3. Residual Plot for Trip Duration on 1% prediction sample

TABLE III
FARE PREDICTION PERFORMANCE

Model	RMSE	MAE	R^2
Linear Regression	5.89	2.89	0.887
Random Forest	5.55	2.52	0.900

The feature importance analysis revealed a notable insight: `rush_hour` (0.281) was significantly more predictive than `trip_duration` (0.003). Since `fare_amount` represents the base metered price (excluding surcharges), this suggests cost of a trip is more strongly correlated with predefined peak traffic windows than with the raw elapsed time of the trip. Effectively, the `rush_hour` feature acts as a proxy for traffic congestion levels, proving more statistically significant than the total duration itself.

D. Trip Classification (Hypothesis 3)

The binary classification task (Short vs. Long trips) was evaluated using ROC-AUC and F1-score to account for the class distribution.

TABLE IV
TRIP CLASSIFICATION PERFORMANCE

Model	Accuracy	F1	ROC-AUC
Logistic Regression	0.860	0.857	0.931
Random Forest	0.878	0.877	0.942

The RF Classifier achieved a near-perfect ROC-AUC of **0.942**. Feature importance for classification was dominated by `trip_distance` (0.838), fol-

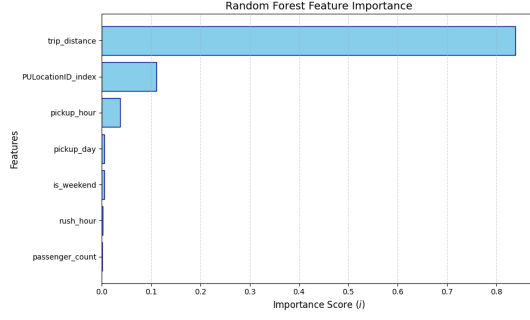


Fig. 4. Feature Importance from RF Trip Classification

lowed by `PULocationID_index` (0.110), confirming that distance and starting location are the most reliable predictors for trip length categorization.

E. Interpretation of Results

The results confirm that while linear models provide a strong baseline for regulated outcomes like fares, Random Forest ensembles are superior for capturing the non-linear "noise" associated with variable factor situations like urban traffic. The high importance of the pickup location across all RF models indicates that spatial context is a critical component of Big Data mobility analytics.

V. CONCLUSION

This project successfully developed a distributed Big Data pipeline to analyze and predict outcomes for 24 million NYC taxi trips. By leveraging the Databricks environment and Spark, we demonstrated that while linear models provide a solid baseline for regulated pricing, ensemble methods like Random Forest are more effective at capturing non-linear complexity.

The findings validated all three research hypotheses. Notably, the importance of the pickup location and rush hour features revealed that the demand and location of a trip is often more predictive of the final fare than the raw trip duration.

Although the models achieved strong predictive performance, further improvements could be achieved through additional data sources. Integrating weather datasets and subway service disruptions could account for some of the remaining variance in trip duration that cannot be explained by taxi data alone. Additionally, hyperparameter tuning via `CrossValidator` could be employed to further refine the Random Forest architectures [3].

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